

Sheet: **DWG-A · Carport A**
Ref.: Proposal **JRG-2026-008**
Date: **September 11, 2026**

PROJECT Two-Vehicle Open Carport — 24'-0" x 25'-0"	LOCATION 18398 131st Trail N, Jupiter Farms, FL	CLIENT Raul	SCALE N.T.S.
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24'-0" CLEAR SPAN

G1-G4: GIRDER
3x12x.188 · 6005A-T61
 24'-0" ONE PIECE, NO SPLICE

SB1-SB2: SIDE BEAM
3x6x3/16 · 6061-T6
 CONTINUOUS, 25'-0"

P1-P8: POST 4x4x1/4
 6061-T6 · 4 PER SIDE
 @ 8'-4" O.C.

FOOTING 2'x2'x3'-0"D
 TYP. 8 — SEE DETAIL 4

SLOPE 1:12

ROOF DECK: 1-1/2" TYPE B · 18 GA · G90 · (9) SHEETS x 27'-0" CONTINUOUS — 1'-0" OVERHANG ALL SIDES — RIBS RUN WITH SLOPE

All columns on the two 25' sides only — both 12' parking bays fully unobstructed. Circled numbers reference connection details below.

The diagram illustrates a bridge deck with three vertical posts. The top surface is labeled 'DECK 1-1/2" TYPE B 18 GA'. An orange line, labeled 'SB 3x6 CONT.' and circled with a '1', runs along the underside of the deck. The posts are labeled 'G3 3x12 (END)'. The clearances from the ground to the underside of the deck are 'CLEAR 9'-0"±' on the left and 'CLEAR 7'-0"±' on the right. The posts are embedded in the ground, with the label 'EMBED 3'-0" TYP.' indicating the typical embedment depth. The ground level is marked 'GR'. A circled '4' is located at the base of the rightmost post.

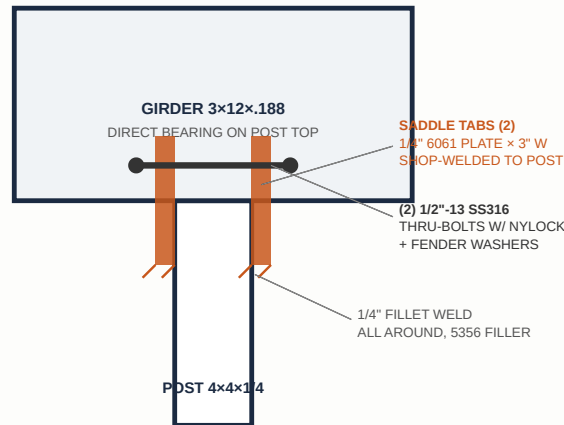
POST HEIGHTS (A.F.G.)

- P1/P5 (front): 9'-0"
- P2/P6: 8'-4"
- P3/P7: 7'-8"
- P4/P8 (rear): 7'-0"

CUT POST STOCK 12'-6"
FIELD-TRIM TO GRADE

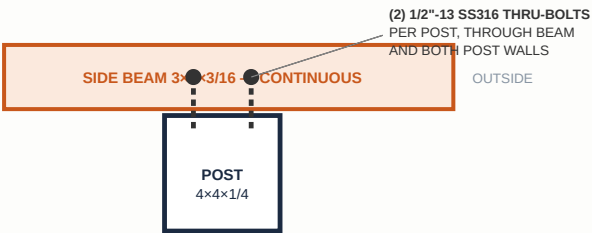
Single slope 1:12 front to rear. Girders level side-to-side; deck ribs drain front to rear.

DETAIL 1 · GIRDER-TO-POST CONNECTION



Girder bears on post top (gravity); saddle tabs + thru-bolts resist wind uplift. Typ. 8 locations.

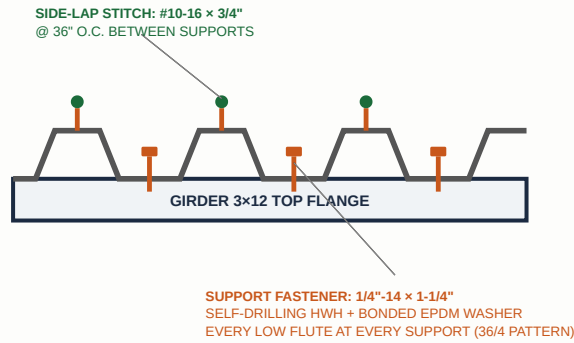
DETAIL 2 · SIDE BEAM AT POST (PLAN SECTION)



SIDE BEAM RUNS ON OUTSIDE FACE OF ALL 4 POSTS, TOP FLUSH WITH GIRDER BEARING — TIES COLUMN LINE, 25'-0" ONE PIECE (SPlice OVER POST PERMITTED, INTERNAL SLEEVE 12" + (4) 3/8" SS BOLTS)

Typ. 8 locations (4 per side). All bolt holes drilled 1/16" oversize, deburred.

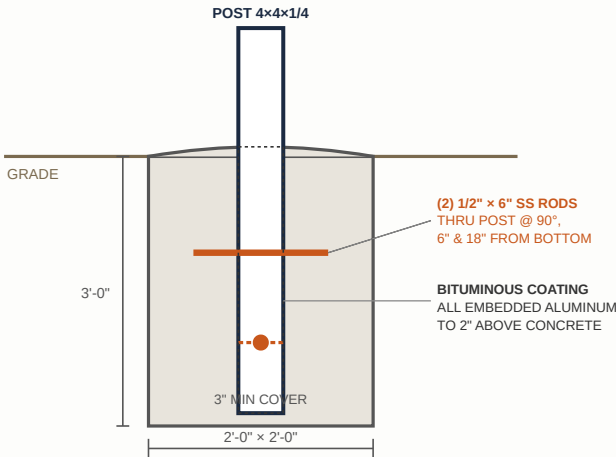
DETAIL 3 · DECK FASTENING (SECTION AT GIRDER)



DECK: 1-1/2" TYPE B · 18 GA · G90 GALVANIZED · SHEETS CONTINUOUS OVER 3 SPANS

Perimeter zone: fastener in EVERY flute. Field: every flute at supports per HVHZ practice.

DETAIL 4 · FOOTING & COLUMN EMBEDMENT



3,000 psi concrete, top crowned to shed water. Post set plumb & braced before pour. Typ. 8.

BILL OF MATERIALS — CARPORT A

Mark	Member	Section	Alloy	Length	Qty
P1–P8	Column	4" × 4" × 1/4" sq. tube	6061-T6	12'-6" (field-trim)	8
G1–G4	Roof girder (clear span)	3" × 12" × 0.188" rect. tube	6005A-T61	24'-0"	4
SB1–SB2	Side beam (continuous)	3" × 6" × 3/16" rect. tube	6061-T6	25'-0"	2
—	Saddle tabs (Detail 1)	1/4" plate, 3" × 6"	6061-T6	—	16
—	Roof deck	1-1/2" Type B, 18 ga, G90 galv.	Steel	27'-0" (3'-0" cov.)	9
—	Footing concrete, 3,000 psi	2'-0" × 2'-0" × 3'-0"	—	0.45 CY each	8

FASTENER SCHEDULE — CARPORT A

Location	Fastener	Qty / Spacing
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Girder to post (Detail 1)	1/2"-13 × 5-1/2" SS316 hex bolt, nylock nut, fender washers	2 per post · 16 total
Side beam to post (Detail 2)	1/2"-13 SS316 thru-bolt, nylock nut, fender washers	2 per post · 16 total
Deck to girders (Detail 3)	1/4"-14 × 1-1/4" self-drilling HWH, bonded EPDM washer	Every low flute at every support
Deck side laps	#10-16 × 3/4" stitch screw	36" o.c. between supports
Footing anchors (Detail 4)	1/2" × 6" SS rod thru post, 2 per post @ 90°	16 rods total

GENERAL NOTES

- All aluminum welding with 5356 filler by certified aluminum welder; 1/4" fillet welds unless noted.
- All bolts, nuts, washers and anchor rods stainless steel Type 316. Bolt holes 1/16" oversize, deburred.
- Galvanized deck bearing directly on aluminum is a compatible (zinc) contact; all deck cut edges touched up with cold-galvanizing compound.
- Embedded aluminum protected with bituminous coating per FBC — no aluminum in direct contact with concrete or soil.
- Layout, post locations and final grades field-verified with owner before excavation. Roof deck overhangs the frame 1'-0" on all four sides. Deck sheet lengths confirmed against as-built frame before ordering.
- Structure designed as open (no walls). Do not enclose sides without review — wind loads change.